Lab#08

Implementation of switch case statement in java

Objective

Implementation of switch case statement in java.

Lab Goals

To provide working insight into the usage of switch case statement in java programming:

a. Switch Statement.

Introduction to switch case statement

In Java, the switch statement is a control flow statement that allows you to select one of many code blocks to execute based on the value of a variable or an expression. It provides a concise way to handle multiple cases and avoid long chains of if-else statements. It has the following syntax.

Syntax

```
switch (expression)
{
  case value1:
  statement1;
  break;
  case value2:
  statement2;
  break;
  case valueN:
  statementN;
  break;
  default:
  statementDefault;
 }
 else
{
  // Executes this block if
  // condition is false
  }
```

Exercise

A. Write a program using switch which choose between following cases:

a: Add two numbers
b: Find power using math function
c: Exit

Program:

```java
import java.util.Scanner;

public class Lab8 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println(x: "Choose an option:");
        System.out.println(x: "a: Add two numbers");
        System.out.println(x: "b: Find power using math function");
        System.out.println(x: "c: Exit");
        System.out.print(s: "Enter choice: ");
        char choice = sc.next().charAt(index: 0);

        switch (choice) {

            case 'a':
                System.out.print(s: "Enter first number  : ");
                double num1 = sc.nextDouble();
                System.out.print(s: "Enter second number : ");
                double num2 = sc.nextDouble();
                System.out.println("Sum = " + (num1 + num2));
                break;

            case 'b':
                System.out.print(s: "Enter base     : ");
                double base = sc.nextDouble();
                System.out.print(s: "Enter exponent : ");
                double exp = sc.nextDouble();
                System.out.println("Result = " + Math.pow(base, exp));
                break;

            case 'c':
                System.out.println(x: "Exiting program. Goodbye!");
                break;

            default:
                System.out.println(x: "Invalid choice!");
        }

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 8> javac Lab8.java
PS C:\Users\Ansharah\Desktop\Java\Lab 8> java Lab8
Choose an option:
a: Add two numbers
b: Find power using math function
c: Exit
Enter choice: a
Enter first number  : 8
Enter second number : 9
Sum = 17.0
PS C:\Users\Ansharah\Desktop\Java\Lab 8>
```

B. Write a program in Java to calculate the total cost of items purchased based on their category. The program should prompt the user to enter the category of the item (1 for electronics, 2 for groceries, and 3 for clothing) and the amount spent on that category. Then, calculate and display the total amount spent on each category separately. Use a switch-case statement to implement this program.

Expected Output:

```
Enter the category of item:
1. Electronics
2. Groceries
3. Clothing
2
Enter amount spent: 75.50
Total spent on Groceries: $75.5
```

Program:

```java
import java.util.Scanner;
public class Lab8 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println(x: "Enter the category of item:");
        System.out.println(x: "1. Electronics");
        System.out.println(x: "2. Groceries");
        System.out.println(x: "3. Clothing");
        int category = sc.nextInt();

        System.out.print(s: "Enter amount spent: ");
        double amount = sc.nextDouble();

        switch (category) {

            case 1:
                System.out.println("Total spent on Electronics: $" + amount);
                break;

            case 2:
                System.out.println("Total spent on Groceries: $" + amount);
                break;

            case 3:
                System.out.println("Total spent on Clothing: $" + amount);
                break;

            default:
                System.out.println(x: "Invalid category!");
        }

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 8> javac Lab8.java
PS C:\Users\Ansharah\Desktop\Java\Lab 8> java Lab8
Enter the category of item:
1. Electronics
2. Groceries
3. Clothing
1
Enter amount spent: 9000
Total spent on Electronics: $9000.0
PS C:\Users\Ansharah\Desktop\Java\Lab 8>
```

C. Write a program in Java to convert a numerical grade into a letter grade. The program should prompt the user to enter a numerical grade between 0 and 100 and then display the corresponding letter grade based on the following grading scale:

- 90-100: A
- 80-89: B
- 70-79: C
- 60-69: D
- Below 60: F

Use a switch-case statement to implement this program.
Hint: grade should be user input for suppose 88 you have taken as user input.
Then divide it **int range = grade / 10;** after that use **range** as an expression in switch Case.
Expected Output:

```
Enter your grade (0-100): 88
Letter Grade: B
```

Program:

```java
import java.util.Scanner;
public class Lab8 {
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter your grade (0-100): ");
        int grade = sc.nextInt();

        // Divide grade by 10 to get range
        int range = grade / 10;

        switch (range) {

            case 10:
            case 9:
                System.out.println(x: "Letter Grade: A");
                break;

            case 8:
                System.out.println(x: "Letter Grade: B");
                break;

            case 7:
                System.out.println(x: "Letter Grade: C");
                break;

            case 6:
                System.out.println(x: "Letter Grade: D");
                break;

            default:
                System.out.println(x: "Letter Grade: F");
        }
        sc.close();
    }
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 8> javac Lab8.java
PS C:\Users\Ansharah\Desktop\Java\Lab 8> java Lab8
Enter your grade (0-100): 34
Letter Grade: F
PS C:\Users\Ansharah\Desktop\Java\Lab 8>
```

D. Write a program in Java to calculate the area of different geometric shapes. The program should prompt the user to choose the shape (1 for circle, 2 for rectangle, and 3 for triangle) and then input the necessary parameters (radius for circle, length and width for rectangle, and base and height for triangle). Finally, calculate and display the area of the chosen shape. Use a switch-case statement to implement this program.

Formula:

1. **Area of Circle: Math.PI * radius * radius**
2. **Area of Rectangle: length * width**
3. **Area of Triangle: 0.5 * base * height**

Expected Output:

```
Choose shape to calculate area:
1. Circle
2. Rectangle
3. Triangle
3
Enter base: 10
Enter height: 5
Area of Triangle: 25.0
```

Program:

```java
import java.util.Scanner;
public class Lab8 {
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println(x: "Choose shape to calculate area:");
        System.out.println(x: "1. Circle");
        System.out.println(x: "2. Rectangle");
        System.out.println(x: "3. Triangle");
        int choice = sc.nextInt();

        switch (choice) {

            case 1:
                System.out.print(s: "Enter radius: ");
                double radius = sc.nextDouble();
                double circleArea = Math.PI * radius * radius;
                System.out.println("Area of Circle: " + circleArea);
                break;

            case 2:
                System.out.print(s: "Enter length: ");
                double length = sc.nextDouble();
                System.out.print(s: "Enter width: ");
                double width = sc.nextDouble();
                double rectArea = length * width;
                System.out.println("Area of Rectangle: " + rectArea);
                break;

            case 3:
                System.out.print(s: "Enter base: ");
                double base = sc.nextDouble();
                System.out.print(s: "Enter height: ");
                double height = sc.nextDouble();
                double triArea = 0.5 * base * height;
                System.out.println("Area of Triangle: " + triArea);
                break;

            default:
                System.out.println(x: "Invalid choice!");
        }
        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 8> javac Lab8.java
PS C:\Users\Ansharah\Desktop\Java\Lab 8> java Lab8
Choose shape to calculate area:
1. Circle
2. Rectangle
3. Triangle
1
Enter radius: 32
Area of Circle: 3216.990877275948
PS C:\Users\Ansharah\Desktop\Java\Lab 8>
```